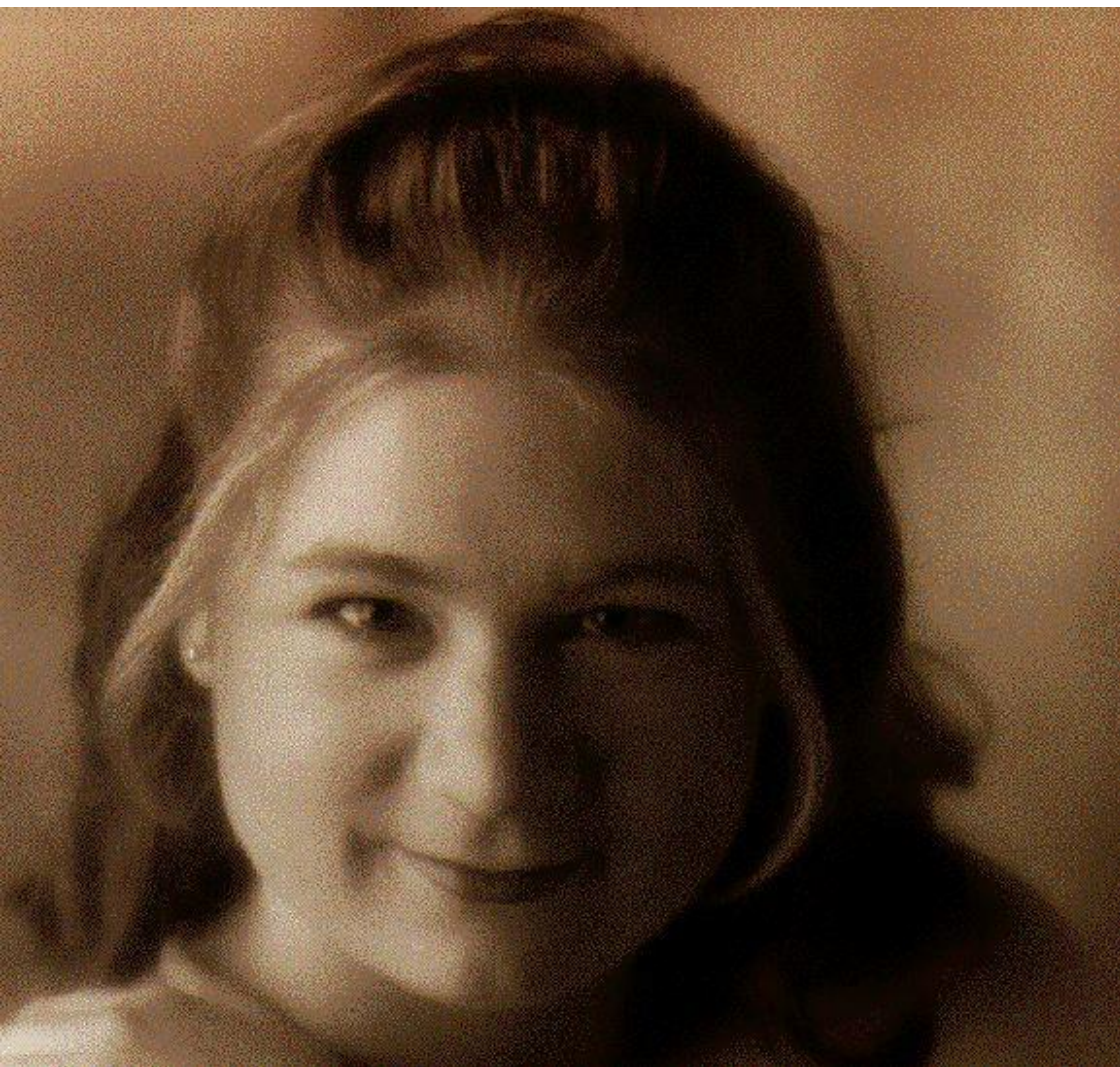


Information Extraction: extracting Named Entities and Relations from text

Based on a presentation by Diana Maynard
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Outline

- Information Extraction applications
- Knowledge Engineering v ML
- Rule-based Named Entity Recognition
- MUSE MUlti-Source Entity recognition
- Research in Information Extraction

IE is not IR

- IR pulls **documents** from large text collections (e.g. “google search” the Web) in response to specific keywords or queries. You analyse the documents.
- IE pulls **facts** and **structured information (Named Entities and Relations)** from the content of large text collections (e.g corpora). You analyse the facts.

IE for Document Access

- With traditional query engines, getting the facts can be hard and slow
 - Where has the Queen visited in the last year?
 - Which places on the East Coast of the US have had cases of West Nile Virus?
- Which search terms would you use to get this kind of information?
- IE would return information in a structured way
- IR would return documents containing the relevant information somewhere (if you were lucky)

IE as an alternative to IR

- IE returns knowledge at a deeper level than IR
- Constructing a database through IE and linking it back to the documents can provide a valuable alternative search tool.
- Even if results are not always accurate, they can be valuable if linked back to the original text

When would you use IE?

- For access to news
 - identify major relations and event types (e.g. within foreign affairs or business news)
- For access to scientific reports
 - identify principal relations of a scientific subfield (e.g. pharmacology, genomics)

Application 1 – HaSIE

- Aims to find out how companies report about health and safety information
- Answers questions such as:
 - “how many members of staff died or had accidents in the last year?”
 - “is there anyone responsible for health and safety”
 - “what measures have been put in place to improve health and safety in the workplace?”

HASIE

- Identification of such information is too time-consuming and arduous to be done manually
- IR systems can't cope with this because they return whole documents, which could be hundreds of pages
- System identifies relevant sections of each document, pulls out sentences about health and safety issues, and populates a database with relevant information

Application 2: KIM

Ontotext's KIM query and results

Search for patterns where

X, is a , which name

and X Y

Y, is a , which name

and Z

Z, is a , which name

Articles (sorted by descending order)

Date & Time (GMT)	Title	Source	Origin
14/11/2002 18:57	Israel: Kibbutz attack suspect arrested [cache]	CNN	
14/11/2002 18:09	Palestinian rivals seek closer ties [cache]	BBC	
14/11/2002 17:47	Kibbutz raid suspect arrested [cache]	BBC	
14/11/2002 17:19	Israelis Capture Palestinian Suspect [cache]	FOX News	
14/11/2002 15:19	Israel Captures Militant Linked to Kibbutz Attack [cache]	Reuters	TULKARM, West Bank
14/11/2002 15:00	Palestinian killed in Nablus; kibbutz "mastermind" surrenders [cache]	AFP	
14/11/2002 13:40	Israel Troops Capture Alleged Gunman [cache]	Guardian	NABLUS, West Bank
14/11/2002 12:47	Israel Arrests Militant Linked to Kibbutz Attack [cache]	Reuters	TULKARM, West Bank
14/11/2002 11:39	Israeli army kills teenager in Nablus [cache]	AFP	
14/11/2002 08:03	Israeli tanks roll into Gaza City after army takes over Nablus [cache]	AFP	
14/11/2002 01:21	Israeli tanks enter Gaza [cache]	BBC	

1-11 of 11 Articles per page:

Application 3: Threat tracker

Threat Tracker - Alias I, Inc - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Huda Ammash 5 Hearts

Saved Searches

"Huda Salih Mahdi Ammash"

Huda Salih Mahdi Ammash

Entities

070 Huda Ammash

SubTrackers

Documents

Huda Ammash - Edit

Excerpt Documents Concordance

Count of how many documents system found

Displaying lead paragraphs from documents mentioning Huda Ammash.

Displaying 1 to 10 of approximately 70 documents.

2.22 - Mrs Anthrax is 18th leader caught - Original - Add to Tracker

Mrs Anthrax is 18th leader caught Mrs Anthrax is 18th leader caught Date: 2003-05-07 04:02:58; Source: The Star (Malaysia) ... With the capture of Huda Salih Mahdi Ammash, a US-trained microbiologist dubbed "Mrs Anthrax", the US military now holds 18 of 55 top Iraqi officials on its most-wanted list. Ammash was the only woman on the list, and was portrayed as the five of hearts on the playing cards of wanted Iraqis produced by the US Defense Department. It described her as a "weapons of mass destruction" scientist. ... made it clear it considered the detention of the 49-year-old woman a major catch, as it struggles to uncover the biological and chemical existence it justified military action against deposed Iraqi leader. Ammash earned a doctorate in microbiology from the University of Baghdad.

Mentions of the Entity Hudda Ammash found by ThreatTrackers

'...' indicates sentences have been removed because they don't mention the Entity

What is Named Entity Recognition?

- Identification of proper names in texts, and their classification into a set of predefined categories of interest
- Persons
- Organisations (companies, government organisations, committees, etc)
- Locations (cities, countries, rivers, etc)
- Date and time expressions
- Various other types as appropriate

Why is NER important

- NER provides a foundation from which to build more complex IE systems
- Relations between NEs can provide tracking, ontological information and scenario building
- Tracking co-reference “Dr Anthrax, Huda, she”
- Ontologies “Person, Event, Country”
- Relations “Dr Head became the new director of Shiny Rockets Corp”

Two kinds of approaches

(to IE, and to other NLP, eg PoS-tagger)

Knowledge Engineering

- “hand-crafted” rule-based system
- developed by experienced knowledge engineers
- makes use of human intuition
- requires only small amount of example training data
- may not require high performance computing
- development can be time consuming
- some changes may require checking entire rule set

Machine Learning

- use statistical language models and machine learning
- developers do not need domain expertise
- requires large amounts of annotated training data
- requires high performance computing: processor, memory
- some changes may require re-annotation of the entire training corpus

Basic Problems in Named Entity Recognition

- Variation of NEs – e.g. John Smith, Mr Smith, John.
- Ambiguity of NE types: John Smith (company vs. person)
 - June (person vs. month)
 - Washington (person vs. location)
 - 1945 (date vs. time)
- Ambiguity between common words and proper nouns, e.g. “may”

Complex problems in NER

- Issues of style, structure, domain, genre etc.
- Punctuation, spelling, spacing, formatting

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- > Tell me more about Leonardo
- > Da Vinci

List lookup approach - baseline

- System that recognises only entities stored in its lists (gazetteers).
- Advantages - Simple, fast, language independent, easy to retarget (just create new lists)
- Disadvantages - collection and maintenance of lists, cannot deal with name variants, cannot resolve ambiguity

Shallow Parsing Approach (internal structure)

- Internal evidence – names often have internal structure. These components can be either stored or guessed, e.g. location:

Cap. Word + {City, Forest, Center, River}
e.g. Sherwood Forest

Cap. Word + {Street, Boulevard, Avenue,
Crescent, Road}
e.g. Portobello Street

Problems with the shallow parsing approach

- Ambiguously capitalised words (first word in sentence)

[All American Bank] vs. All [State Police]

- Semantic ambiguity

"John F. Kennedy" = airport (location)

"Philip Morris" = organisation

- Structural ambiguity

[Cable and Wireless] VS.

[Microsoft] and [Dell]

[Center for Computational Linguistics] vs.

message from [City Hospital] for [John Smith]

Shallow Parsing Approach with Context

- Use of context-based patterns is helpful in ambiguous cases
- "David Walton" and "Goldman Sachs" are indistinguishable
- But with the phrase "David Walton of Goldman Sachs" and the Person entity "David Walton" recognised, we can use the pattern "[Person] of [Organization]" to identify "Goldman Sachs" correctly.

Identification of Contextual Information

- Use KWIC concordance eg SketchEngine to find windows of context around entities
- Search for repeated contextual patterns of entities
- Manually post-edit list of patterns, and incorporate useful patterns into new rules
- Repeat with new entities

Examples of semantic patterns with entity types and relations

- [PERSON] earns [MONEY]
- [PERSON] joined [ORGANIZATION]
- [PERSON] left [ORGANIZATION]
- [PERSON] joined [ORGANIZATION] as [JOBTITLE]
- [ORGANIZATION]'s [JOBTITLE] [PERSON]
- [ORGANIZATION] [JOBTITLE] [PERSON]
- the [ORGANIZATION] [JOBTITLE]
- part of the [ORGANIZATION]
- [ORGANIZATION] headquarters in [LOCATION]
- price of [ORGANIZATION]
- sale of [ORGANIZATION]
- investors in [ORGANIZATION]
- [ORGANIZATION] is worth [MONEY]
- [JOBTITLE] [PERSON]
- [PERSON], [JOBTITLE]

MUSE – Multi-Source Entity Recognition

- An IE system developed within GATE General Architecture for Text Engineering
<https://gate.ac.uk/>
- Performs NE and coreference on different text types and genres
- Uses knowledge engineering approach with hand-crafted rules
- Performance rivals that of machine learning methods

MUSE Modules

- Document format and genre analysis
- Tokenisation
- Sentence splitting
- POS tagging (Verb, Adjective, Noun, ProperNoun)
- Gazetteer lookup (NE dictionaries)
- Semantic grammar ([PERSON] earns [MONEY])
- Name coreference (Mr Smith, John)
- Pronoun coreference (John, he)

Multilingual MUSE

- MUSE has been adapted to deal with different languages: English, French, German, Romanian, Bulgarian, Russian, Cebuano, Hindi, Chinese, Arabic, ...
- Separation of language-dependent and language-independent modules and sub-modules makes adaptation easier.

IE in Surprise Languages

- Adaptation to an unknown language in a very short timespan
- Cebuano:
 - Latin script, capitalisation, words are spaced
 - Few resources and little work already done
 - Medium difficulty
- Hindi:
 - Non-Latin script, different encodings used, no capitalisation, words are spaced
 - Many resources available
 - Medium difficulty

What does multilingual NER require?

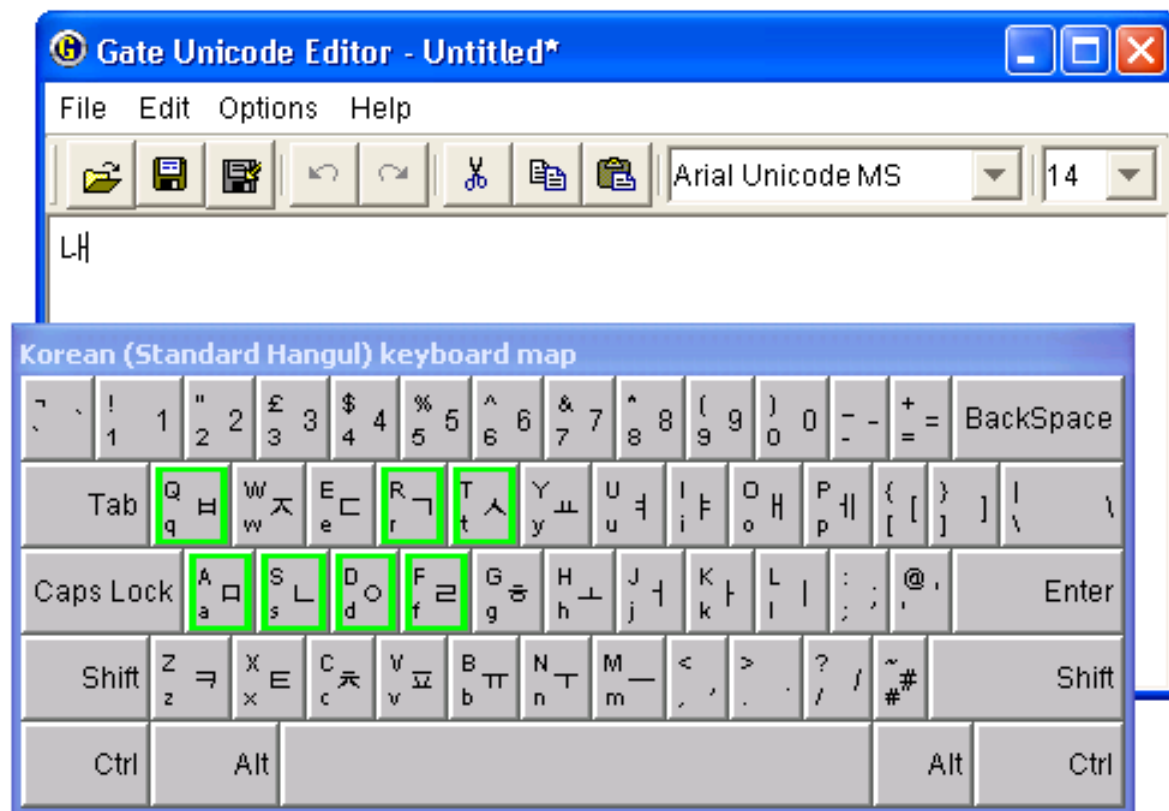
- Extensive support for non-Latin scripts and text encodings, including conversion utilities
 - Takes up to 2/3 of the effort
- Bilingual dictionaries
- Annotated corpus for evaluation
- Internet resources for gazetteer list collection (e.g., phone books, yellow pages, bi-lingual pages)

Editing Multilingual Data

GATE Unicode Kit (GUK)

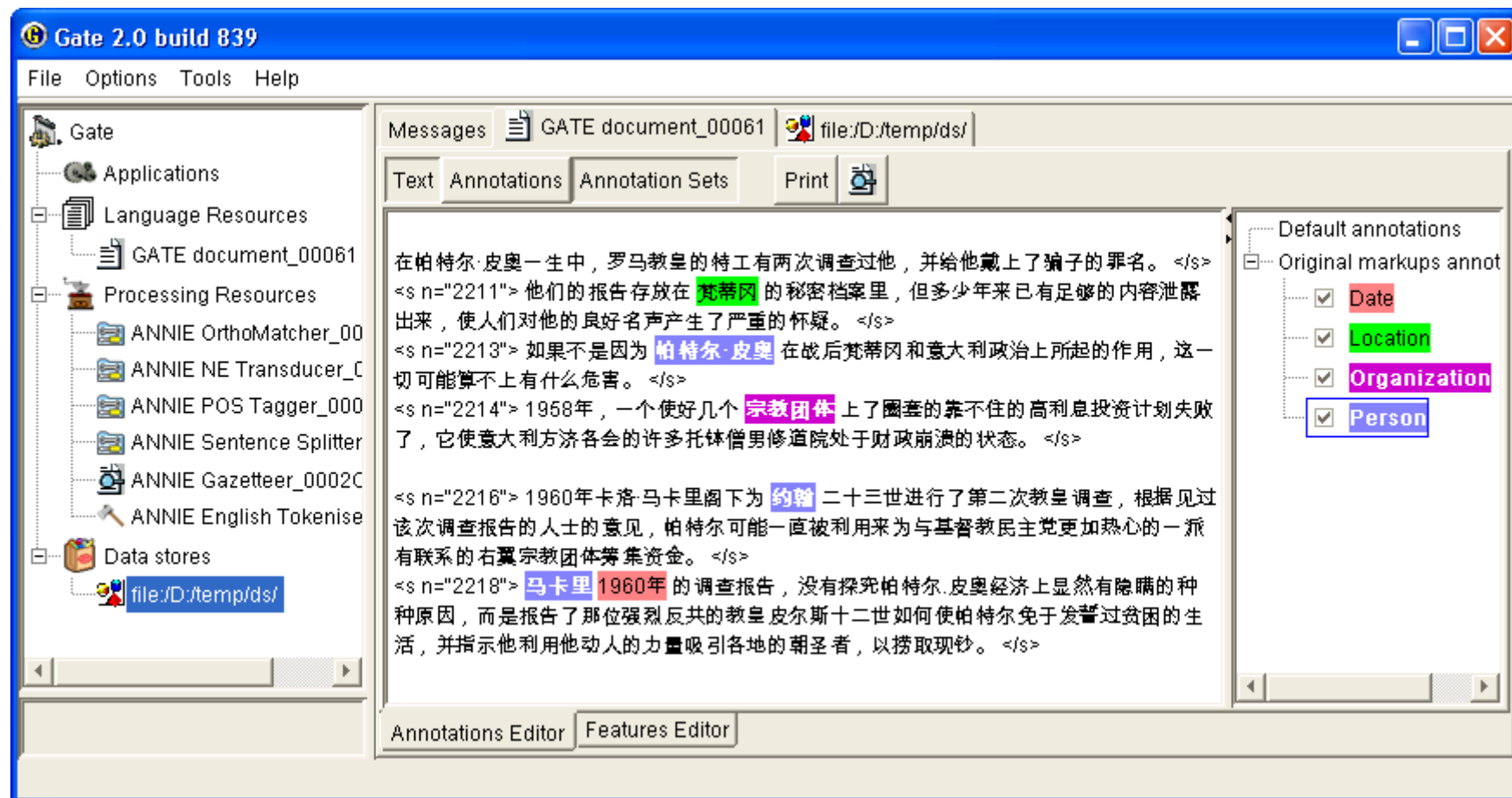
Complements Java's facilities

- Support for defining Input Methods (IMs)
- currently 30 IMs for 17 languages
- Pluggable in other applications (e.g. JEdit)



Processing Multilingual Data

All processing, visualisation and editing tools use GUK



Research in Information Extraction

- Tools for semantic web
- Need for IE in bioinformatics and medicine
- IE for finance, eg IE from news to predict share prices
- Cross fertilisation of IE and IR , eg. Question Answering
- **Relations** between entities
- see Jurafsky and Martin textbook

Summary

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